Roll No. : ....

MID-SEMESTER EXAMINATION 2025

Name of the Program: B.Tech

Name of the Course: **Soft Computing**

Time: 90 Minutes

Semester: 8^{th}

Course Code: TCS-821

Maximum Marks: 50

Note:

i. Answer all the questions by choosing any one of the sub-questions.

ii. Each question carries 10 marks

	(10 Marks)	CO1
Q1		
(a)	Explain the differences between biological neural networks and artificial neural networks.	
	OR	
(b)	Illustrate and explain the McCulloch-Pitts neuron model with an example.	CO1
Q2	(10 Marks)	
(a)	What are the limitations of single-layer perceptron networks? How do they overcome multi-layer perceptron?	
	OR	
(b)	Discuss the applications of soft computing in industry.	
Q3	(10 Marks)	CO1
(a)	Explain different types of activation functions used in artificial neural networks.	
	OR	
(b)	What is the significance of weight initialization in ANN training?	
Q4	(10 Marks)	CO2
(a)	Describe the applications of neural networks in real-world scenarios.	
	OR	
(b)	Explain the significance of fuzzy logic in artificial intelligence and machine learning.	
Q5	(10 Marks)	
(a)	Explain fuzzy composition and its importance in fuzzy logic systems.	
	OR	
(b)	How is Hebbian learning applied in artificial neural networks? Describe a real-world application of Hebbian learning in neural networks.	CO2